VrikkaTwin · system architecture

A kidney digital twin for heatwaves and sick days

"Heat loads the gun; a sick day or a painkiller pulls the trigger."

1 TWO DATA STREAMS

Static EHR
- Age, sex, city, occupation
- BP · diabetes · CKD · baseline eGFR
- Daily medicines
- **Gene:** CYP2C9 (painkiller clearance)

Wearables & IoT
- **Morning weight** (smart scale)
- **Resting heart rate**, steps, sleep
- Home BP cuff
- No device: **ASHA weekly visit**

Weather
- Real hourly ERA5 weather
- **7 Indian cities**, 2024-2026
- Noisy 72-h forecast

Check-ins & labs
- Symptoms, tablets, ORS
- In English · हिन्दी · ಕನ್ನಡ · తెలుగు
- Lab creatinine, potassium

2 SAFETY FIRST

Checked first

Red flags
Confusion, fits, ≥ 40 °C
→ 🚑 **call 108 / 112**
before any AI

Integrity gate
Plausible value
Fresh, not replayed
Device-signed

Not designed for
Under 18, pregnancy,
eGFR < 15 / dialysis

3 DIGITAL TWIN

Personal kidney twin

Heat exposure
Weather + daily routine → **sweat per hour** (ISO 7933)

Kalman filter, every 08:00
Estimates hidden **body water** and **creatinine** from weight + heart rate, and learns **this person's sweat rate**

72-hour simulation
200 possible futures on the forecast, scored with the KDIGO kidney-injury rule

Physiology: autoregulation · triple whammy · exertion · haemoconcentration

4 PREDICT & EXPLAIN

Fast clock · 72 h
- Hybrid **LightGBM model M**
- Twin outputs + **25 clinical features**
- Calibrated % + **top-3 reasons**

Alert level
Green · Amber · Red · Confirmed
- Red ≤ 0.5 per 100 patient-days
- Sent to patient, family, doctor

What-if simulator
- ① Swap painkiller → ② ORS · work hours
- ③ Pause medicine (doctor decides)
- Guardrails: ORS, K^+, heart failure

Slow clock · 10 years
- Bayesian eGFR trend
- **Ranges only**, never one number
- KFRE where valid

5 TRUST LAYER

Every read

Consent
Signed, ABDM-style
Revoke in one click

Audit log
Hash-chained
Signed checkpoints
Tamper-evident

Standards
FHIR R4 export
DPDP Act 2023 map

6 USERS

Doctor
- Ranked triage + reasons
- Kidney Weather by city
- **Order home kidney test**

Patient & family
- 4 languages + audio
- 🚑 **Ambulance button**
- 💊 Medicine list, consent

ASHA worker
- Tier-0 form in Telugu
- No smartwatch needed
- Red flags + 🚑

Lab
- Home sample, ₹189 test
- Result flows **back into the twin**

EVIDENCE

Synthetic training world
15,000 patients: 10,000 train (2024) · 5,000 separate test (2025, 2026). Built with a **different kidney model** + hidden causes, so the AI cannot copy itself.

Real public data checks
MIMIC-IV & **eICU** ICU creatinine · **Tawam** 491 CKD patients · **PMData** Fitbit noise · **UCI CKD** lab values

Published evidence
Triple whammy (BMJ, 487,372 patients) · EMPA-KIDNEY · heat-worker studies · CPIC genetics · KDIGO · ICMR-INDIAB

Research prototype · synthetic & open data only · not a medical device